

1. The short run is a period of time in which
 - A) nothing the firm does can be altered.
 - B) the amount of output is fixed.
 - C) prices and wages are fixed.
 - D) the quantities of some resources the firm uses are fixed.
2. An example of a variable resource in the short run is
 - A) an employee.
 - B) capital equipment.
 - C) land.
 - D) a building.
3. A cost that has already been made and cannot be recovered is called a sunk cost.
4. The long run is a time frame in which
 - A) the quantities of all resources are fixed.
 - B) the quantities of all resources can be varied.
 - C) the quantities of some resources are fixed and the quantities of other resources can be varied.
 - D) all costs are sunk costs
5. In the long run, a firm can vary
 - A) its capital but not its labor.
 - B) its labor but not its capital.
 - C) both its labor and its capital.
 - D) neither its labor nor its capital.

6.

Labour (workers per day)	Total Product (units per day)	Marginal Product	Average Product
0	0	0	0
1	2	2	2
2	8		
3	12		
4	15		
5	16	1	

- a. In the above table, the total product that is produced when the firm employs four workers is
 - A) 8
 - B) 15
 - C) 3.75
 - D) 3

b. In the above table, the marginal product of the third worker is

- A) 2
- B) 1
- C) 4
- D) 3

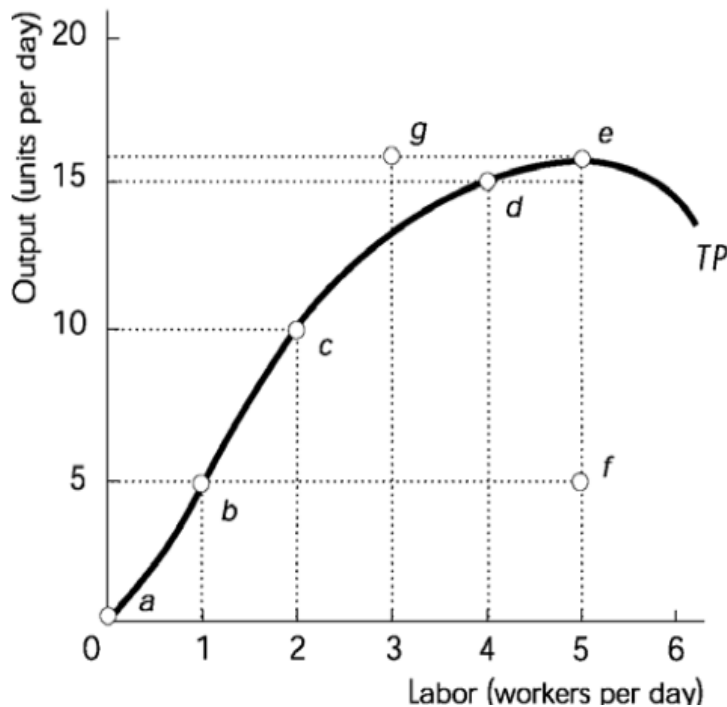
Ans b. MP of 3rd worker = $(\text{TP of 3 labour} - \text{TP of 2 labour}) / (3 - 2) = (12 - 8) / (3 - 2) = 4$

c. In the above table, the marginal product of the fourth worker is

- A) 6
- B) 3
- C) 1
- D) 4

Ans c. MP of 4th worker = $(15 - 12) / (4 - 3) = 3$

7.



In the figure above, the marginal product of the second worker is

- A) 10 units.
- B) 2 units.
- C) 5 units.
- D) 1 unit.

$(\text{TP of 2 labour} - \text{TP of 1 labour}) / (2 - 1) = (10 - 5) / (2 - 1) = 5$ units

8. A company could produce 99 units of a good for \$316 or produce 100 units of the same good for \$320. The marginal cost of the 100th unit

- A) is \$320.
- B) is \$3.20.

C) is \$4.00 : $\text{MC of the 100th unit} = (\text{TC of 100 units} - \text{TC of 99 units}) / (100 - 99) = (320 - 316) / 1 = 4$

9.

Cost schedule			
Labor (workers)	Output (units per day)	Total variable cost (dollars)	Total cost (dollars)
0	0	0	30
1	3	20	50
2	8	40	70
3	12	60	90
4	14	80	110
5	15	100	130

a. In the above table, the total fixed cost is

- A) \$30.
- B) \$50.
- C) \$20.
- D) \$0.

Ans a. At output = 0,

$$TC = TFC + TVC$$

$$TFC = TC - TVC = 30 - 0 = 30$$

b. In the above table, when output increases from 8 to 12 units, the marginal cost is

- A) \$1.20.
- B) \$15.00.
- C) \$5.00.
- D) \$2.00.

Ans b. $MC = \text{change in } TC / \text{change in } O = (90 - 70) / (12 - 8) = \5

c. In the above table, the average fixed cost of producing 15 units of output is

- A) \$6.66.
- B) \$0.50.
- C) \$2.00.
- D) \$8.66.

Ans c. $AFC = TFC / O = 30 / 15 = \2

d. In the above table, the average variable cost of producing 14 units of output is

- A) \$0.175.
- B) \$7.86.
- C) \$5.71.
- D) \$10.00.

Ans d. $AVC = TVC / O = 80 / 14 = \5.71

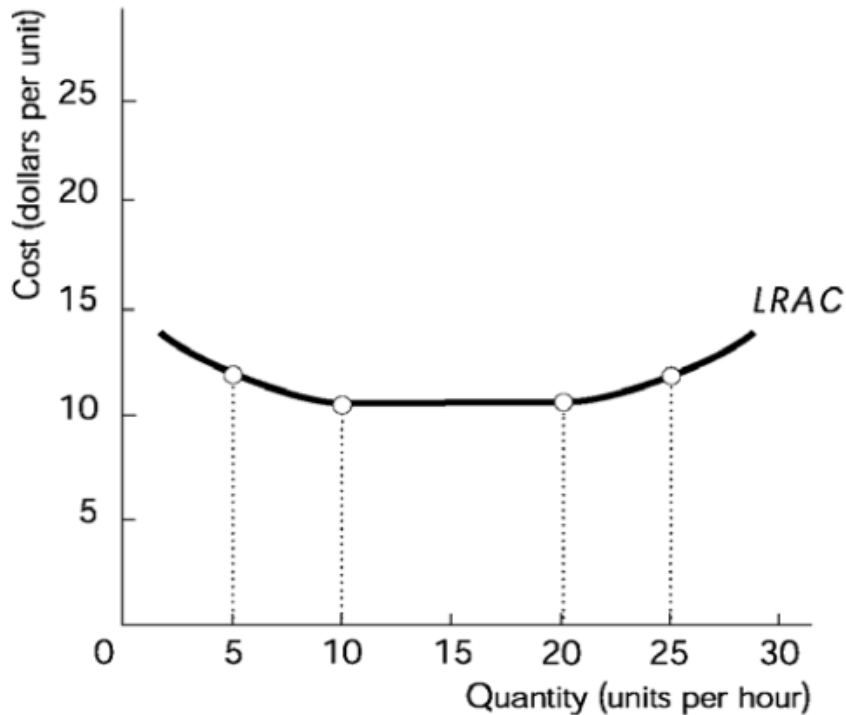
- e. In the above table, the average total cost of producing 14 units of output is
- A) \$5.71.
 - B) \$7.00.
 - C) \$7.86.
 - D) \$6.75.

Ans e. $ATC = TC/O = 110/14 = \$7.86$

10. When long-run average costs decrease as output increases, there are economies of scale.

11. When long-run average costs increase as output increases, there are diseconomies of scale.

12.



- a. In the above figure, the long-run average cost curve exhibits economies of scale
 - A) between 5 and 10 units per hour. [cost decreasing]
 - B) between 20 and 25 units per hour.
 - C) between 10 and 20 units per hour.
 - D) along the entire curve.
- b. In the above figure, the long-run average cost curve exhibits constant returns to scale
 - A) between 5 and 10 units per hour.
 - B) between 20 and 25 units per hour.
 - C) between 10 and 20 units per hour. [cost constant]
 - D) along the entire curve.
- c. In the above figure, the long-run average cost curve exhibits diseconomies of scale
 - A) between 10 and 20 units per hour.
 - B) between 5 and 10 units per hour.
 - C) between 20 and 25 units per hour. [cost increasing]
 - D) along the entire curve.

13. The table below represents a coffee company's total cost of producing coffee. Using the provided information, find the values of FC, VC, MC, AFC, AVC, and ATC.

Quantity	TC= FC+VC	FC	VC	MC	AFC	AVC	ATC
0	50	50	0	-	-	-	-
1	55	50	5	$(55-50)=5$	$(50/1)=50$	$(5/1)=5$	$(50+5)=55$
2	58	50	8	$(58-55)=3$	$(50/2)=25$	$(8/2)=4$	$(25+4)=29$
3	59	50	9	$(59-58)=1$	$(50/3)=16.7$	$(9/3)=3$	$(16.7+3)=19.7$
4	64	50	14	$(64-59)=5$	$(50/4)=12.5$	$(14/4)=3.5$	$(12.5+3.5)=16$
5	70	50	20	$(70-64)=6$	$(50/5)=10$	$(20/5)=4$	$(10+4)=14$

★ If it's not asked in the question to find out AFC, then when solving for ATC, instead of going for $ATC=AFC+AVC$, use $ATC=TC/Q$. You are supposed to get the same result. You can solve ATC for this same problem using the stated method.

14. The table below represents information about a coffee company's total product. Using the provided information, find the values of MP & AP.

Number of Labour	Total Product (thousand per year)	Marginal Product	Average Product
0	0	-	-
1	10	$(10-0)/1=10$	$(10/1)=10$
2	28	$(28-10)/1=18$	$(28/2)=14$
3	54	$(54-28)/1=26$	$(54/3)=18$
4	76	$(76-54)/1=22$	$(76/4)=19$
5	90	$(90-76)/1=14$	$(90/5)=18$

Note: All these problems have been taken from external sources.